Final Project Report

Meal Planner App for the Elderly (MealCare)

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1. Reflection on the use of generative AI

Generative AI was used as support, not as a substitute for design or ownership of the work. Different tools being used on different phases for example perplexity helped in research based work such as identifying UI/UX standards for elderly peoples and nutrition considerations. For implementation, ChatGPT and DeepSeek being used for clarifying ideas and concepts for example React patterns and debugging, sketching approaches, and suggesting tests.

Over time, the use of AI become more controlled. Instead of fully relying, it was used to:

- Clarify concepts
- Identify possible causes of errors and loop holes
- Improve code structure

2. Problems encountered and how they were resolved

2.1. Barcode matching

Products did not matched with scanned products due to naming issue some times API return same product with different name formats

Cause: Matching that was too strict effectively exact comparison after a simple trim.

Resolution: improved matching by cleaning names first like making text to lowercase, removing specials characters and symbols instead of directly compare shopping item with scanned item.

2.2. State management in React

Consistency of data such meal planning, shopping, user's profile on several screens because these are going to be used whole way in app .

Cause: Early data states was split into different ways without a single source of truth that made data hard to access.

Resolution: Lifting state up make globally accessible under system and concentrating meal-plan logic in custom hooks for example **AppContext.jsx** with context where several areas need the same data. That reduced duplication and made errors easier to trace.

2.3. Supabase integration

Connecting Supabase auth, profiles, and persisted data involved asynchronized behavior.

Cause: Limited prior experience with backend-as-a-service and end-to-end async flows.

Resolution: Working in small steps call Supabase in isolated environment, then connect the UI, and checking responses before assuming the UI was wrong. That cut down on **mystery bugs** that were hard to catch and debug.

2.4. UI design for elderly users

On earl stages the design layouts were closer to a generic app than to elderly specific such as cognitive load and accessibility.

Cause: Feature density and information overload before usability was checked.

Resolution: Larger controls and type, simpler navigation, reduce cognitive load Including follow-up fixes with color schemes and text formats. The aim was a calm, legible interface aligned with older users and with accessibility expectations.

3. What I achieved

- User profile with dietary, health and budget preferences and etc. .
- Weekly meal planning, meal selection, and Progress tracking of meal Planning.
- Smart shopping list generation from the meal plan, market based aisle sections.
- Barcode lookup **Open Food Facts** with matching items where possible, and marking items bought/add to shopping list.
- Caregiver support: account roles and a caregiving area with linking between elderly who need assistance .
- Rule-based nutrition and filtering for example diet and health constraints in recipe search, implemented as explicit logic rather than a black box recommender .

The system is deployable as a client application with a Supabase backend, consistent with the chosen architecture.

4. What I did not achieve

- Highly personalized **recommendation engine** beyond the rule based nutrition and filtering that is in the app
- Deeper caregiver: for example full care-team workflows, analytics dashboard, role-based access are in place, but not a full care-management platform.
- Integration with supermarket checkout or delivery APIs.

5. How the software was tested

Testing was carried out in multiple ways using manual test cases and automated tools.

I created own test cases to check core features such as meal selection, shopping list generation, user profile, barcode scanning. This approach ensures that each features work according to requirement.

In addition, several tools being used for testing such as LightHouse to evaluate performance, accessibility and usability, This helps to take attention on these areas before deployment phase.

Accessibility was also tested using WCAG guidelines **(Web Content Accessibility Guidelines)** which focuses on aspects such as text readability, color contrast and font sizes to ensure application meets the Elderly UI standards.

6. What I learned

Technical: Deeper hand-on-practice with React structure, hooks, and async integration; using Supabase for auth and data storage and implementing a real device feature such as barcode integration with device camera access.

Design: Usability for elderly peoples is not just **the same as usual but with details**, it is pace, clarity, and error tolerance.

AI: Generative tools are useful for learning and drafts, but integration and evaluation are always my responsibility.

7. What I would do differently

- Sketch data flow and state ownership earlier to reduce refactors.
- Build structured plan and split features into possible iterations rather than start working randomly.
- Run structured usability checks earlier .

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I have cited the sources of all quotations, paraphrases, summaries of information, tables, diagrams or other material, including software and other electronic media in which intellectual property rights may reside.

I have cited clearly, exhaustively and adequately in my source code, code generated by Gen AI tools.

I understand that I can only get credit for work that I have done (e.g. understanding other code, interfacing my code with it, modifying it, etc.), but you will not be credited with the third party code itself.

I have provided a complete bibliography of all works and sources used in the preparation of this submission.

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